Venkata Lakshmi Parimala Pasupuleti

Software Engineer · LLM Evaluation Infrastructure · Eval-Driven Development · Python · Reward Modeling · CI/CD · Observability · FastAPI · PyTorch · HuggingFace

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PROFESSIONAL SUMMARY

Software Engineer with 8+ years building production-grade distributed systems, now deeply focused on AI evaluation infrastructure. Built AgentProbe: open-source eval toolkit with a CI-integrated 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-judge scoring, golden dataset regression harness, and distributed model routing across OpenAI/Anthropic/Groq/Ollama — the exact "evals first" infrastructure this role describes. Built Leasel: 8-agent AI platform evaluated via a custom pass@k harness (93.6% precision, 87.9% F1, pass@1 81.7% / pass@3 88.3% across 1,240 test cases) where LLM reasons and deterministic logic makes every regulated decision — eval-driven design from the ground up. At Entrata, instrumented production services with OpenTelemetry, Prometheus, and Grafana for 100K+ MAU at 99.5% uptime. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, evaluation pipelines, and data APIs surfacing insights from raw business data into application-facing reporting layers; debugging reliability issues across the application stack.

George Mason University Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source prediction models (Python, R) across 190+ countries achieving R^2=0.977; full pipeline from data collection through hypothesis testing, confidence intervals, and validation. Applied Isolation Forest to automate quality checks across 3 research pipelines — surfacing anomalies that manual review missed.
- Fine-tuned a custom classification model using HuggingFace Transformers + PyTorch (transformer architecture, training loop, eval methodology); deployed inference API; presented model design, training, and evaluation results to faculty. Delivered technical workshops to 600+ students across Python and database design.

Entrata (Enterprise SaaS / Real Estate Technology) Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Instrumented production services with observability tooling** (OpenTelemetry, Prometheus, Grafana) — built monitoring, alerting, and operational dashboards for mission-critical data pipelines serving 100K+ monthly active users; reduced debugging time by 40%; maintained bug count below 10 for 6+ consecutive months. Owned through production incidents with durable root-cause fixes, not workarounds.
- **Built fault-tolerant event-driven distributed systems** (RabbitMQ) processing thousands of account state changes — retry logic, idempotency guarantees, CI/CD (GitHub Actions), Docker containerization; 50% scalability improvement, 30% deployment time reduction across 120+ Agile sprints. Owned full SDLC: design → test → deploy → operate → incident response.
- **Owned multi-tenant AWS-hosted infrastructure** (S3, EC2, RDS) — schema design for 100K+ housing units; optimized 200+ SQL queries (30% performance gain) via execution plan analysis and indexing; 100% PHI compliance on AWS S3 migration with zero security incidents. Mentored 5+ engineers; 300+ peer code reviews.

Entrata India, Pune Jul 2017 – Sep 2018

Associate Software Engineer

- Built backend modules (Java, Python, PostgreSQL); led AWS S3 cloud migration; managed data pipelines across hundreds of thousands of records with output validation and quality gates.

PROJECTS

AgentProbe: AI Evaluation Infrastructure & Observability Toolkit EVAL-DRIVEN CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · OpenTelemetry | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-judge scoring; CI-integrated regression harness with golden datasets and reproducible test conditions; distributed model routing across OpenAI/Anthropic/Groq/Ollama; trace-level observability. Built to catch AI quality regressions before they reach production users — the same infrastructure this role is tasked with building for Siri.

Leasel: 8-Agent AI Platform — Eval-Driven Design + HITL PASS@K HARNESS LLM-AS-JUDGE

Python · LangGraph · FastAPI · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Evaluated via a custom pass@k harness with multi-trial sampling, regression gate, and LLM-as-judge: 93.6% precision, 87.9% F1, pass@1 81.7%, pass@3 88.3% across 1,240 lease PDFs (5 US states + federal). Core design principle enforced in code: LLM extracts and reasons; deterministic logic or human makes every final regulated decision. Fault-tolerant HITL escalation with conditional routing and durable state management.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), Java (Spring Boot), JavaScript, TypeScript, SQL, PHP, Bash, R — FastAPI, Django, React.js, REST APIs

AI Evaluation & LLMOps: LLM Evaluation, LLM-as-Judge, Eval-Driven Development, Reward Modeling, RLHF, RLAIF, Golden Datasets, Regression Harnesses, Pass@k, Scoring Methodology, Ground Truth, Trajectory Analysis, Outcome Analysis, Inter-Rater Agreement, Hallucination Detection, Offline + Online Evaluation, AgentProbe, LangChain, LangGraph, RAG Pipelines, ChromaDB, HITL, Agent Tracing, OpenAI API, Anthropic API

ML & Data Science: PyTorch, HuggingFace Transformers, Transformer Architectures, scikit-learn, Isolation Forest, Model Versioning, Confidence Intervals, Hypothesis Testing, Feature Engineering, NumPy, Pandas, Tableau, Jupyter

Cloud, Infra & Observability: AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Docker, Kubernetes, CI/CD (GitHub Actions), Terraform, OpenTelemetry, Prometheus, Grafana, Distributed Systems, Microservices, Multi-Tenant Architecture, PostgreSQL, Redis